Predictive Fault Detection Report

LSTM-based anomaly detection and failure classification

Generated 15 Apr 2026, 23:05

Dataset	predictive_maintenance.csv	Input Shape	(10, 6)	Classes	6

Run Overview

Field	Value
Generated	2026-04-15 23:05:30
Dataset	predictive_maintenance.csv
Input Shape	(10, 6)
Classes	Heat Dissipation Failure, No Failure, Overstrain Failure, Power Failure, Random Failures, Tool Wear Failure

Run Settings

Setting	Value
Epochs	60
Batch Size	64
Time Steps	10
Simulator	Disabled
Simulation Steps	200

Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Binary LSTM	0.9819	0.6000	0.1875	0.2857	0.9302
Multiclass LSTM	0.9626	0.9830	0.9626	0.9711	N/A

Binary Accuracy	0.982	Multiclass Accuracy	0.963	Elapsed	0.9 min

How the System Works

The pipeline first cleans the industrial sensor data, converts it into time-series windows, balances the training folds safely after sequencing, and then trains multiple architectures. The binary model decides whether the machine is in a fault or no-fault state, the multiclass model identifies the specific failure type, and the RUL model estimates how many steps remain before failure.

During evaluation, the project reports accuracy, F1, ROC-AUC, PR-AUC, MCC, calibration, and a model comparison table. The simulator then uses the saved models to show live risk, the predicted failure type, and the estimated remaining useful life.

Fault vs No-Fault Identification

Binary classification is the first decision layer: when the model predicts No Fault, the machine is treated as healthy; when it predicts Fault, the system escalates to the multiclass classifier to name the specific failure type. This makes the output easier to interpret in a maintenance setting and gives a direct link between sensor patterns and machine state.

In the current run, the binary model achieved 0.982 accuracy with an F1 of 0.286, while the multiclass model achieved 0.963 accuracy and 0.971 weighted F1. The comparison shows how each architecture balances detection quality and calibration across the fault classes.

Model Comparison

Task	Architecture	Accuracy	F1	ROC-AUC	PR-AUC	MCC	Training Time (
Binary	LSTM	0.9807	0.0	0.9025	0.3518	0.0	0.0
Multiclass	LSTM	0.8135	0.8861	0.9618	0.9898	0.2544	0.0
Binary	CNN+LSTM	0.9819	0.2857	0.9302	0.3605	0.3288	0.0
Multiclass	CNN+LSTM	0.9626	0.9711	0.9457	0.9905	0.4517	0.0
Binary	Transformer	0.9807	0.0	0.8718	0.1973	0.0	0.0
Multiclass	Transformer	0.892	0.9322	0.9628	0.9913	0.3103	0.0

Binary LSTM — Visualizations

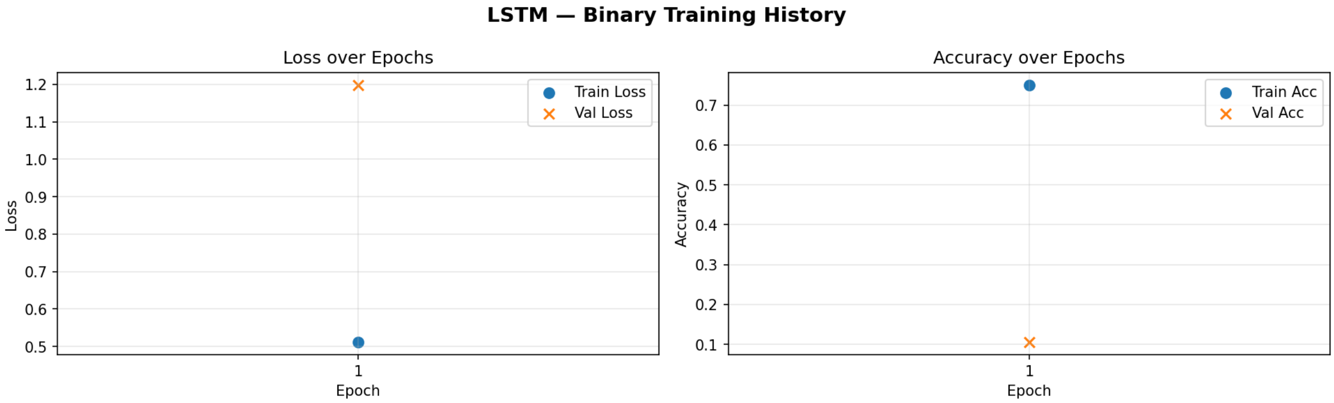

Figure 1 — Binary LSTM: Training Loss & Accuracy

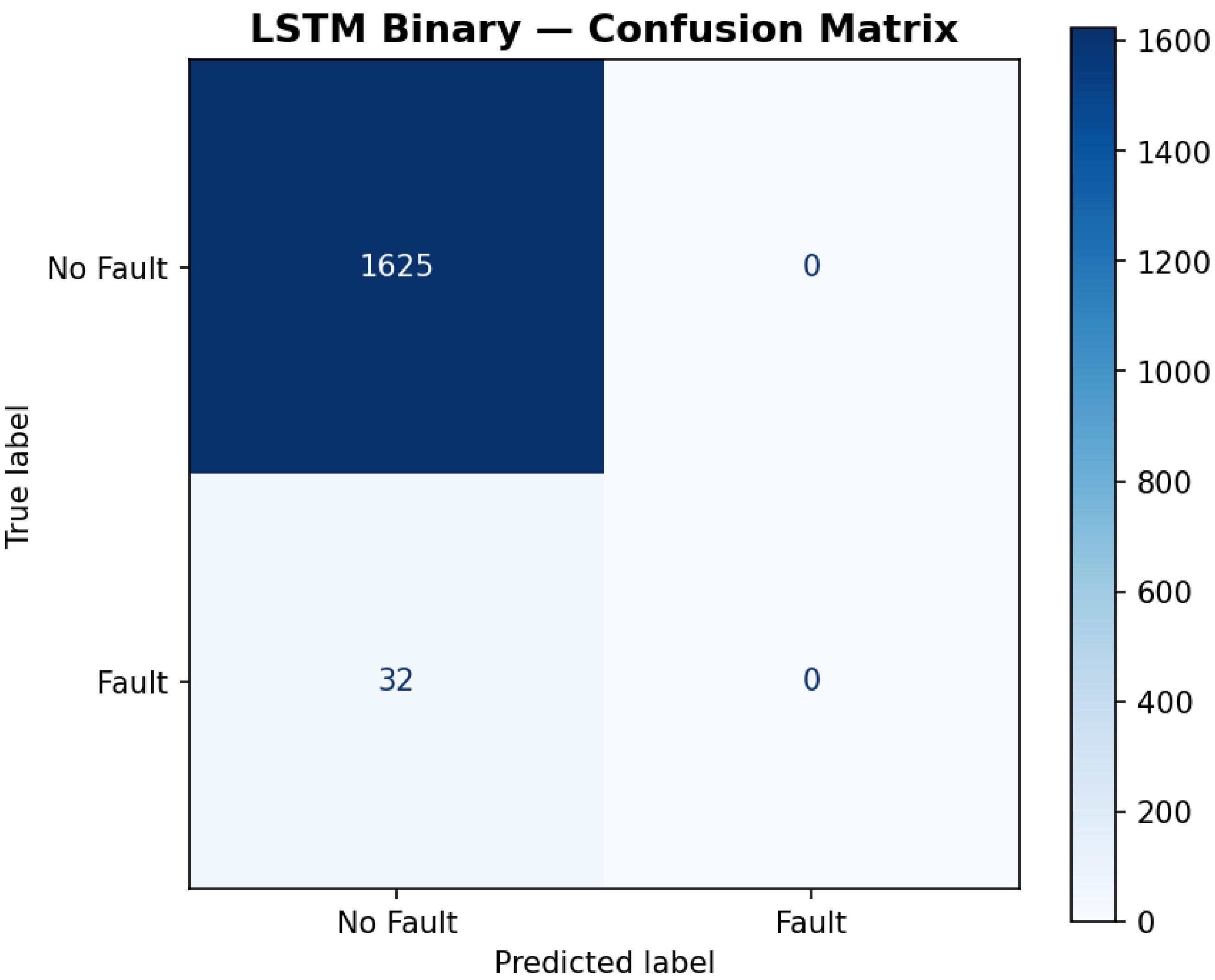

Figure 2 — Binary LSTM: Confusion Matrix

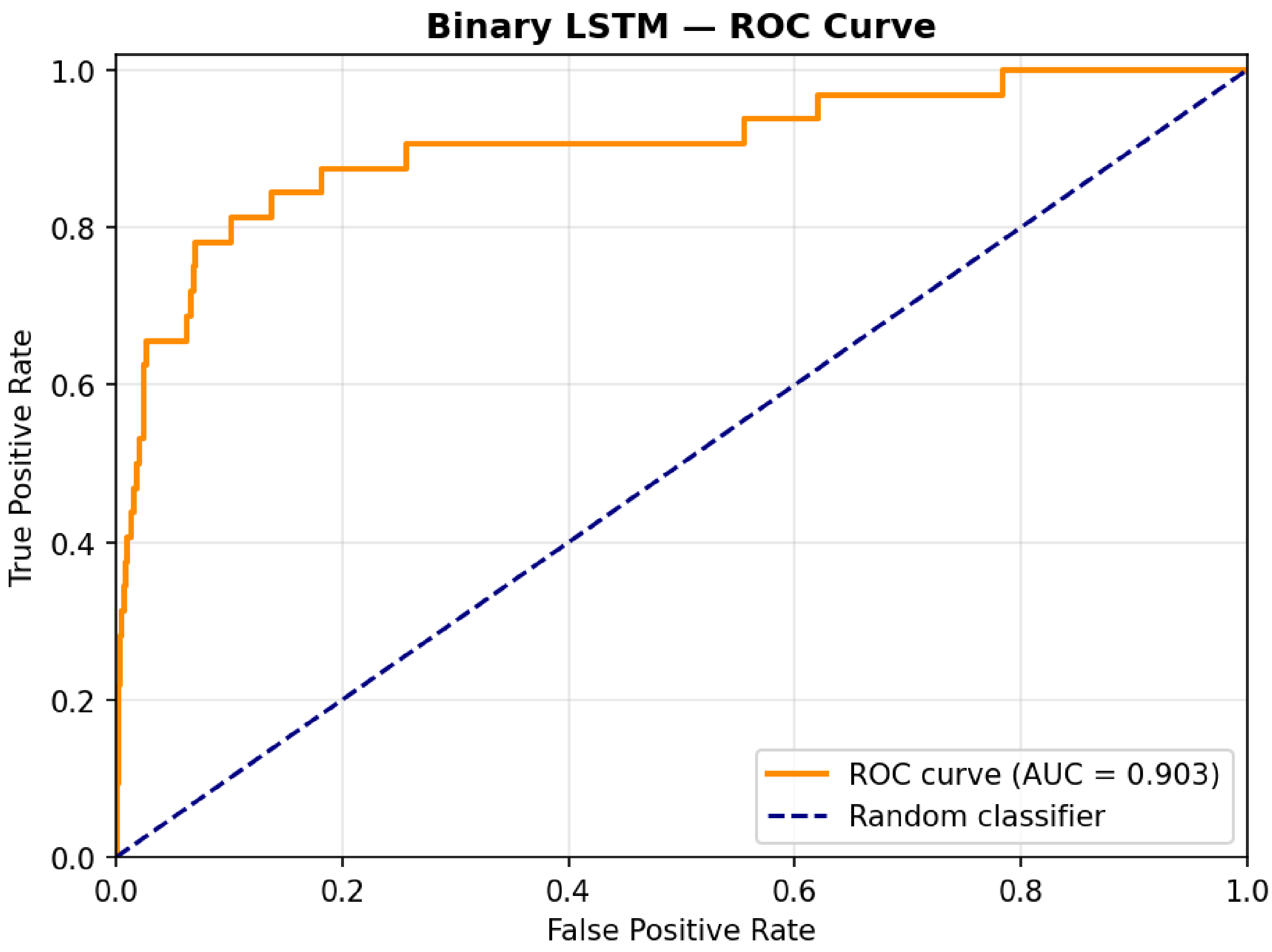

Figure 3 — Binary LSTM: ROC Curve

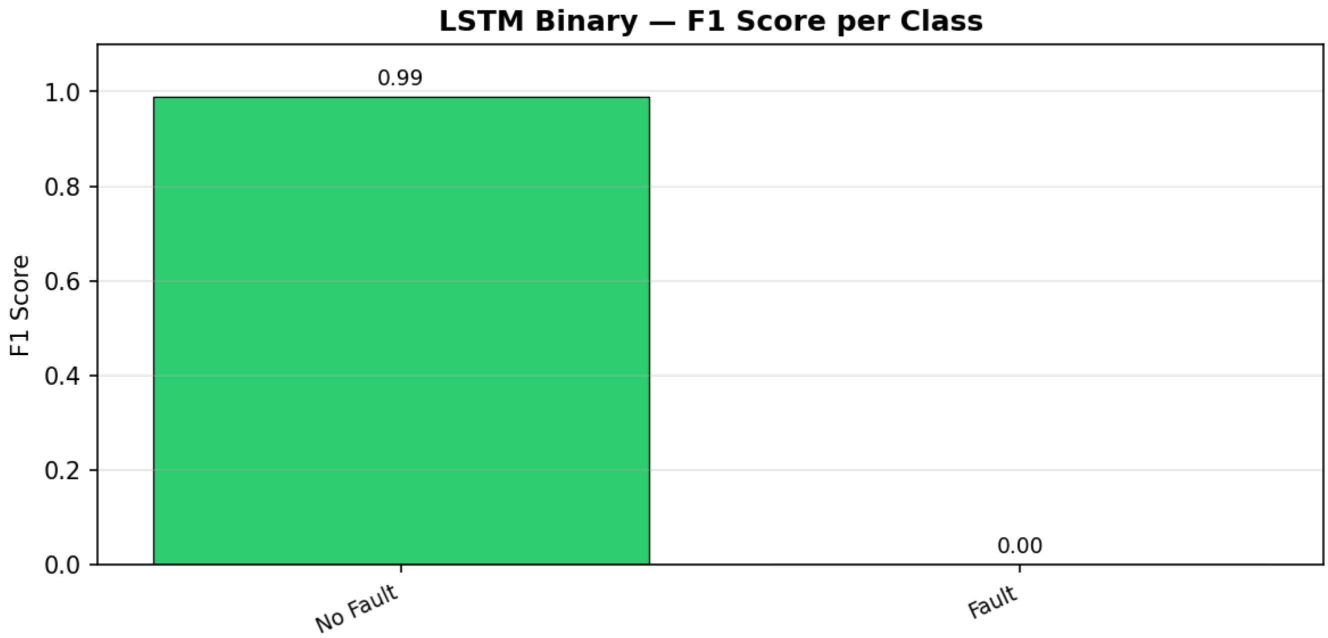

Figure 4 — Binary LSTM: F1 Score per Class

Multiclass LSTM — Visualizations

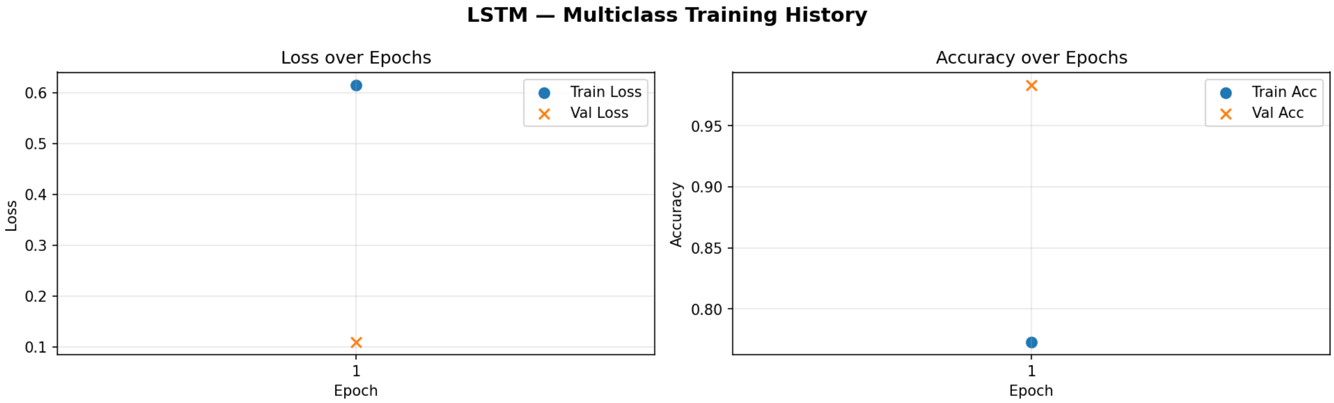

Figure 5 — Multiclass LSTM: Training Loss & Accuracy

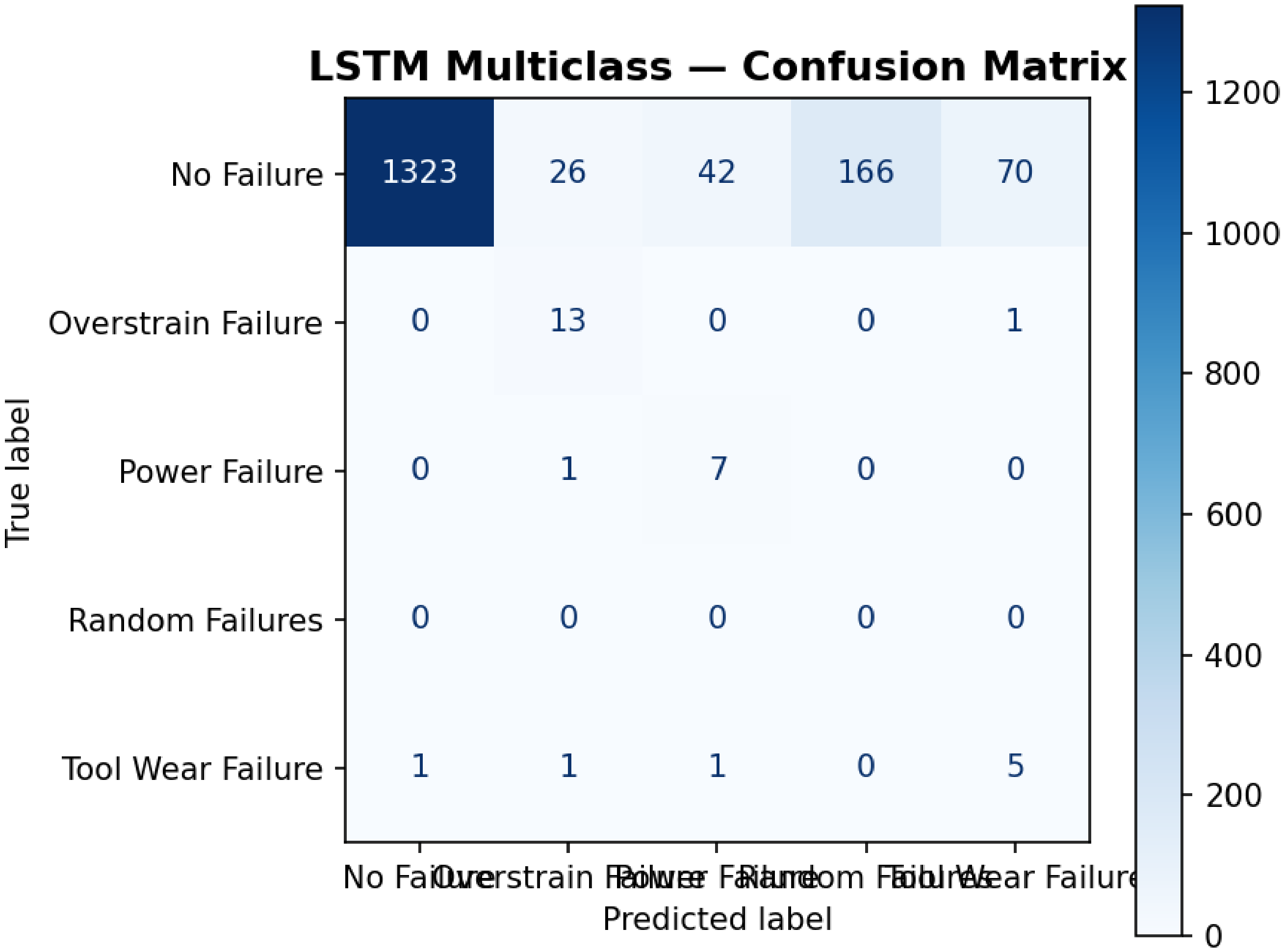

Figure 6 — Multiclass LSTM: Confusion Matrix

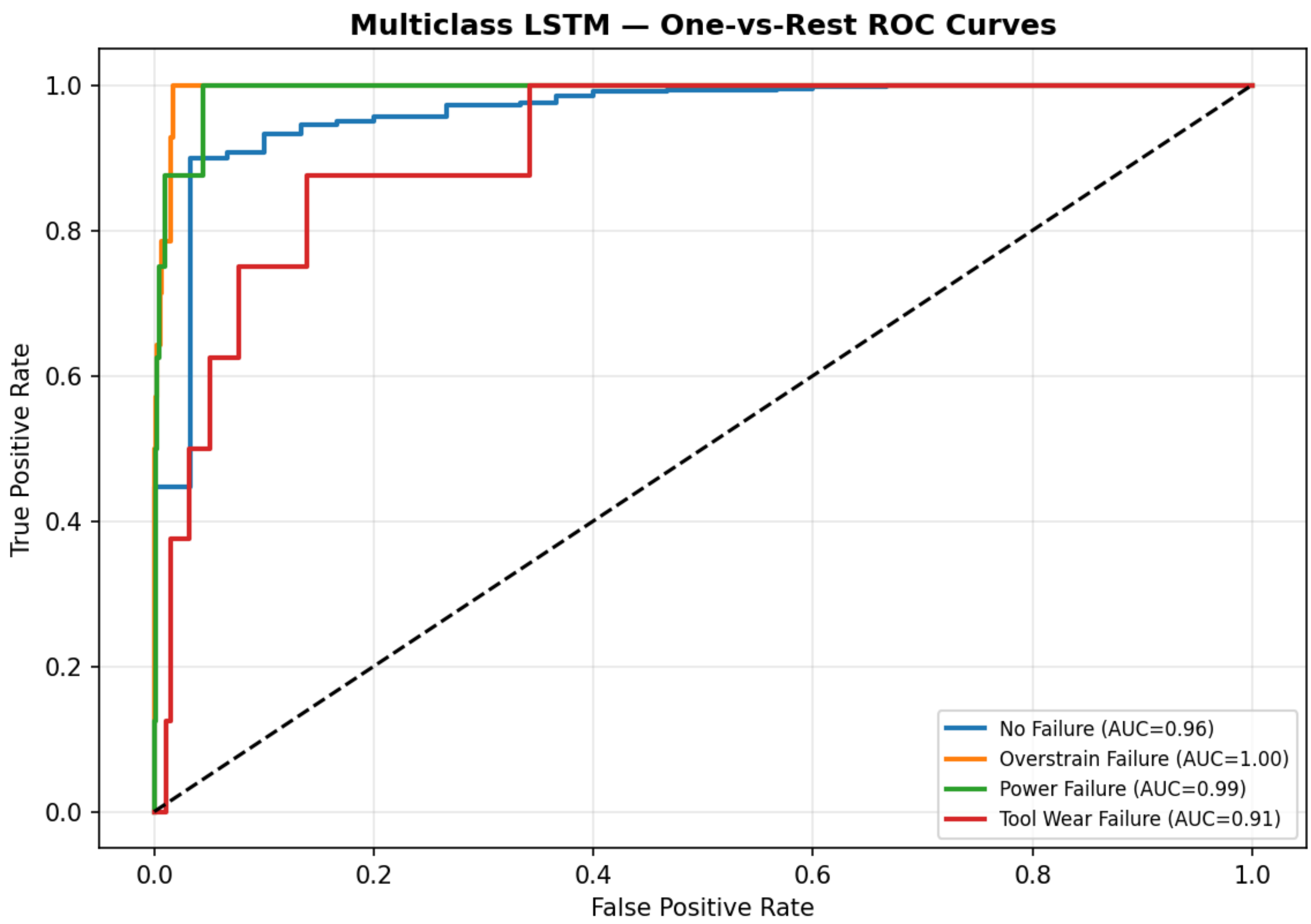

Figure 7 — Multiclass LSTM: One-vs-Rest ROC Curves

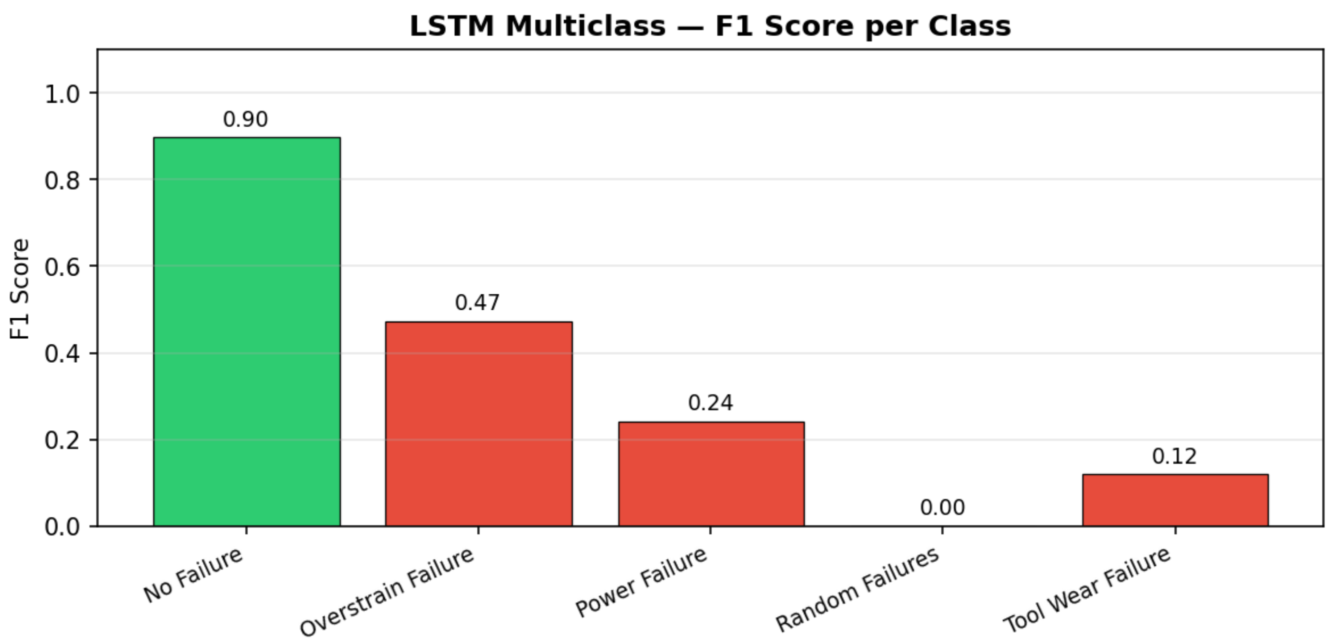

Figure 8 — Multiclass LSTM: F1 Score per Class

Generated Files

Status	File	Description
OK	01_Binary_LSTM_Training_History.png	Binary LSTM training curves

OK	02_Binary_LSTM_Confusion_Matrix.png	Binary confusion matrix
OK	03_Binary_LSTM_ROC_Curve.png	Binary ROC curve
OK	04_Binary_LSTM_F1_Scores.png	Binary F1 scores per class
OK	05_Multiclass_LSTM_Training_History.png	Multiclass LSTM training curves
OK	06_Multiclass_LSTM_Confusion_Matrix.png	Multiclass confusion matrix
OK	07_Multiclass_LSTM_ROC_Curves.png	Multiclass ROC curves (one-vs-rest)
OK	08_Multiclass_LSTM_F1_Scores.png	Multiclass F1 scores per class
OK	results_summary.csv	Combined metrics summary
SKIPPED	simulation_log.csv	Robot simulator step-by-step log
OK	Report.pdf	Full PDF report with embedded plots

Total elapsed time: 0.9 min